PARSHANT BALWARIA

Full-Stack Software Engineer · Geospatial Systems

SUMMARY

Full-stack software engineer with 3+ years at Ramboll and MapMyIndia, building Python/FastAPI and React/TypeScript systems on PostgreSQL/PostGIS, with geospatial systems as a specialty. Built 2 production platforms and deployed 10+ internal tools. Recent work includes real-time locking for a 500 GB+ shared production map database, a ~60× speed-up on the NYSERDA TENs district energy platform, and a serverless AWS module for VU.CITY's 3D digital city platform.

EXPERIENCE

Ramboll — Assistant Digital Product Engineer III **Oct 2025 – Present**
Gurgaon, Haryana, India

- **Renewable Energy Investment Platform:** took over technical ownership mid-project after the lead engineer left, owning software decisions and client requirements end to end. Wind, solar and Power-to-X site screening combining GIS suitability analysis, infrastructure routing, financial modelling (NPV, LCOE/LCOF, CAPEX/OPEX) and automated investment reporting, behind Azure SSO and enterprise RBAC. *Python, FastAPI, PostgreSQL/PostGIS, React, TypeScript, Azure.*
- **NYSERDA TENs (district energy economic screening):** contributed two modules to an international project about two years into development, delivered July 2026. Helped achieve a ~60× overall performance improvement, cutting single processing runs from ~48 hours to under an hour, by removing a heavy ArcPy / file-geodatabase workflow and a single merged layer with hundreds of columns and millions of rows, and eliminating redundant loops and repeated calculations. The project team received a Ramboll certificate of appreciation.
- **VU.CITY platform status messaging (Aug 2026 – Nov 2026):** built the error, warning and success messaging module for VU.CITY's core 3D digital city platform with their data team, over mesh, raster, GeoJSON and Shapefile data in PostGIS. Designed the module and reviewed and tested the output of an agentic coding workflow. *Python, AWS Lambda, Terraform, PostGIS.*
- Migrated geospatial pipelines from ArcGIS/ArcPy file geodatabases to GeoPandas/GeoPackage, and integrated ArcGIS Pro, Online and Enterprise with backend APIs and dashboards.

MapMyIndia (CE Info Systems) — Software Associate II → Software Engineer III **Nov 2023 – Sep 2025**
Okhla, New Delhi, India *Promoted Jan 2025*

- **Collaborative map editing and locking:** built real-time locking for a shared production PostGIS database holding 500 GB+ of all-India map data, edited concurrently by 200+ editors. Each edit locks only the affected, adjacent and related features, replacing manual coordination of who could edit which area. *PostGIS, FastAPI, React, PyQGIS, CI/CD.*
- Led the move of production and testing teams from proprietary software to a centralized QGIS + PostgreSQL environment: a custom QGIS plugin to connect and manage layers, end-to-end logging, user-level security, and a FastAPI activity dashboard for user actions, system logs and security events.
- Built an offline MQTT messaging service that keeps remote GIS operators in sync over poor connectivity, and a QGIS plugin toolkit that automates cartography tasks and enforces data standards.
- Core developer on a MapStore + QGIS Server + PostGIS deployment, optimising rendering speed and security.
- Set up internal infrastructure: a unified QGIS plugin repository, an offline PyPI server, Docker deployments, GitLab CI/CD and Sphinx documentation.
- Received a second, unrequested increment about three months after promotion, for delivery quality.

MapMyIndia — Software Engineer Trainee (Internship) **Aug 2022 – Oct 2022**

- Built QGIS plugins for filtering and updating map data.

SELECTED PERSONAL PROJECTS

- **NexClasses** — https://nexclasses.site: production SaaS for a K-12 coaching centre. Turborepo monorepo with a Node/Express + Prisma API on PostgreSQL 16 and a React 18 + Vite frontend; JWT auth with Admin/Teacher roles, student lifecycle, batch scheduling with overlap detection, fee billing and an installable PWA with web push. Deployed by GitHub Actions to AWS EC2 (Docker Compose, Nginx, Let's Encrypt), monitored with Sentry and UptimeRobot.
- **Builder's Ledger** — https://buildersledger.online: discrete scheduling simulation of three parallel upgrade queues for a mobile strategy game (fan project), comparing nine sequencing strategies under income and prerequisite limits and reporting the bottleneck queue. A data pipeline packs 1,271 PNGs (~47 MB) into a ~320 KB WebP sprite pack inside a single self-contained page. *Node.js, data pipeline, WebP.*
- **Notch:** self-hosted, offline-first expense tracker PWA. Writes go to IndexedDB (Dexie) first and sync later; daily AES-256 encrypted pg_dump backups with tiered retention and automated test restores. *FastAPI, PostgreSQL 16, TypeScript, Docker, Nginx.*
- **Home Server Monitor:** event-driven Python monitor (~0.2% CPU, <100 MB RAM) for a 24/7 Docker home server, tracking dual internet lines, power, containers and disk health. No open ports: FastAPI dashboard behind Cloudflare Tunnel/Access, with a Cloudflare Worker for outage alerts.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, HTML, CSS
Backend:	FastAPI, Django, Node.js / Express, Prisma, REST APIs, MQTT, JWT auth
Databases:	PostgreSQL + PostGIS, MySQL, SQLite
GIS:	QGIS / PyQGIS, QGIS Server, MapStore, ArcGIS Pro / Online / Enterprise, ArcPy, GeoPandas, Leaflet
Frontend:	React, Vite, Tailwind CSS, Bootstrap, PWA
Cloud & DevOps:	Azure, AWS (Lambda, EC2), Terraform, Docker, Nginx, GitHub Actions, GitLab CI, Linux, Cloudflare
Practices:	Agentic coding, CI/CD, Sphinx documentation, Data structures & algorithms

ACHIEVEMENTS

- Ramboll certificate of appreciation (2026) to the NYSERDA TENs Project Team for dedication, teamwork and perseverance.
- Led Tech and Python initiatives at GirlScript for a year, running virtual Python classes and Hacktoberfest events.
- Completed the 30-day Google Cloud program with certification.

EDUCATION

M.Tech, Computer Science & Engineering (Distance) — Maharshi Dayanand University, Rohtak 2025 – 2027 (pursuing)

B.Tech, Computer Science & Engineering — UIET, Kurukshetra University (CGPA 7.26) 2019 – 2023

XII (HBSE) — Nobel Dev Public School · X (HBSE) — Govt. Sr. Sec. School

CONTACT

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